Salahaddin University-Erbil
College of Computer Science and IT
Stage 2 Cs & IT

Introduction to Web Design-Lec. 1

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Tuesday: IT- 8:30-10:30/ Hall 3
CS- 10:30 -12:30/ Hall 16

What you will gain from this course:

- Understand how websites work (browser ↔ server).
- Build simple, semantic pages with HTML5.
- Style and layout pages with CSS.
- Make pages responsive, accessible, and performant.

Course map & learning outcomes

Learning outcomes (after this lecture, you should be able to…)

Describe the basic web architecture; write valid HTML; apply core CSS rules; create responsive layouts;

Lecture structure

- Part 1: What “web design” means & how the web works
- Part 2: HTML (structure + semantics)
- Part 3: CSS (style + layout)
- Part 4: Responsive design + accessibility
- Part 5: Basics of JavaScript, tools, performance, deployment

The Internet vs the World Wide Web

Key definitions

Internet: a global network of networks (physical + routing).
Web (WWW): an information system that uses URLs + HTTP(S) to deliver documents/resources via browsers.

- A website is a collection of web pages and resources under a domain (e.g., example.com).
- A web page is a single document (usually HTML) that can reference CSS, JavaScript, images, and fonts.
- A browser (Chrome/Firefox/Edge) renders HTML/CSS and executes JavaScript.
- A web server stores and sends resources when requested (Apache, Node, etc.).
- Protocol: an agreed set of rules (HTTP for web, TCP/IP for transport).

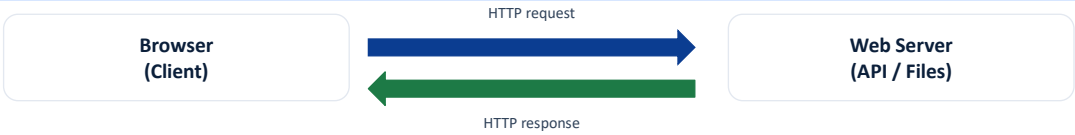
How the web works: client–server + HTTP

- Client: the user's device + browser that sends requests.
- Server: a machine/service that responds with resources (HTML/CSS/JS/images/data).
- HTTP request: method (GET/POST), URL, headers, and sometimes a body.
- HTTP response: status code (200, 404, 500…), headers, and a body (content).

•**200** = "OK, here you go."
•**404** = "I can't find it."
•**500** = "Something broke on my side."

Important terms you see in exams

URL: address of a resource. DNS: translates domain names to IP addresses.
HTTPS: HTTP over TLS (encryption + integrity). Cookie/session: state across requests.



Web design vs web development

Web design (focus: user experience)

- Concerned with how a site looks and feels.
- Includes layout, typography, color, spacing, visual hierarchy.
- Creates wireframes, mockups, UI components, and style guides.
- Must consider accessibility and usability.
- Example: designing a homepage that clearly guides the user to “Apply” or “Contact”.

Web development (focus: functionality)

- Concerned with building the site/app to work correctly.
- Front-end: HTML/CSS/JS in the browser.
- Back-end: server logic, databases, APIs, authentication.
- Must consider security, performance, and maintainability.
- Example: building a login system and storing users in a database.

HTML + CSS + JavaScript (the front-end trio)

- HTML (HyperText Markup Language): defines structure and meaning (semantics).
- CSS (Cascading Style Sheets): controls presentation (colors, spacing, layout).
- JavaScript: adds behavior and interactivity (events, dynamic content).

Browser rendering pipeline (high-level)
1) Parse HTML → DOM tree. 2) Parse CSS → CSSOM. 3) Combine → Render tree.
4) Layout (sizes/positions). 5) Paint (pixels). 6) Composite (layers).

Essential tools and a basic workflow

Tools you should know

- Code editor:notepad, notepad++ VS Code, bracket,…etc.
- Browser DevTools: inspect HTML/CSS, console logs, network requests.

Typical workflow

- 1) Plan: requirements + content + target users.
- 2) Design: wireframe → mockup → components.
- 3) Build: HTML skeleton → CSS styling → layout → responsive.
- 4) Enhance: JavaScript interactions (as needed).
- 5) Test: different screens/browsers, accessibility, performance.
- 6) Deploy: host + domain + HTTPS.

HTML basics: elements, tags, attributes

Definition

- HTML stands for Hypertext Markup Language
- It is used to create and structure web pages
- HTML describes the content of a webpage, not its behaviour or design

Why HTML Is Important

- Foundation of all websites
- Works together with CSS and JavaScript
- Every web browser understands HTML

HTML vs Programming Languages

- HTML: Markup language (structure only)
- Programming languages: Logic and decision making

Example: HTML shows a button, JavaScript controls what happens

HTML files use .html or .htm extension
index.html is the main page of a website
HTML files are plain text files

HTML basics: elements, tags, attributes

Definition

- An HTML element is a piece of the document (e.g., a paragraph, heading, link). Tags mark where an element starts/ends.
- Tags are keywords enclosed in angle brackets <>. Most tags come in pairs: opening and closing
- Closing tags use a slash </ >

- Opening tag: Closing tag:
- Content sits inside the element.
- Some elements are void (no closing tag): <img>, <br>, <meta>.
- Attributes: provide additional information about elements written inside the opening tag.
- Example: href, src, alt
- Attributes are key="value": <a href="...">
- class and id are used for styling and scripting hooks.

```
<h1 class="title">My Portfolio</h1>
<p>Welcome to my website.</p>
<a href="projects.html">See projects</a>
<img src="me.jpg" alt="My Personal Photo">
```

Mini example

HTML document structure (must-know)

- <!DOCTYPE html> tells the browser to use HTML5 standards mode.
- **<html> is the root element (often includes lang="en").**
- **<head> contains metadata (not visible content).**
- **<title> page title in the browser tab.**
- **<meta charset="utf-8"> for correct text encoding.**
- **<meta name="viewport" ...> for responsive layout on phones.**
- **<body> contains the visible page content.**
  - <h1> to <h6> define headings
  - <h1> is the largest heading
  - <h6> is the smallest heading
  - tag defines a paragraph
  - Used for normal text content
  - Automatically adds spacing before and after

Canonical HTML5 skeleton

```
<!DOCTYPE html>
<html lang="en">

  <head>
    <meta charset="utf-8" />
    <meta name="viewport"
     content="width=device-width, initial-
     scale=1" />
    <title>Example Page</title>
    <link rel="stylesheet" href="styles.css" />
  </head>

  <body>
    <h1>Hello, Web!</h1>
    <p> Write your paragraph here</p>
  </body>

</html>
```

Semantic HTML5: structure with meaning

Definition

Semantic HTML uses elements that describe their role (meaning) in the page (e.g., <nav>, <main>, <article>) instead of using only generic blocks.

Why semantics matter

- Accessibility: screen readers understand structure better.
- SEO: search engines better understand content importance.
- Maintainability: code is easier to read and update.
- Consistency: standard patterns across sites.

Common semantic elements

- <header>: top area of a page/section.
- <nav>: major navigation links.
- <main>: unique main content (one per page).
- <section>: thematic grouping (with heading).
- <article>: self-contained piece (post, news).
- <footer>: bottom area (copyright, contacts).

Links & navigation: moving between pages

Definition

A hyperlink (<a>) connects one resource to another using a URL. Navigation is a structured set of links that helps users move around a site.

- Absolute URL: includes protocol + domain (https://example.com/page).
- Relative URL: path from current file (./about.html, /images/logo.png).
- Target behaviors: target="_blank" opens a new tab.
- Good UX: navigation labels should match user goals (Home, Courses, Contact).
- Accessibility: links should be descriptive (“Download syllabus PDF”).

Example navigation

```
<nav>
  <a href="index.html">Home</a>
  <a href="courses.html">Courses</a>
  <a href="contact.html">Contact</a>
</nav>

<!-- external link -->
<a href="https://developer.mozilla.org/"
        target="_blank“>
  MDN Web Docs
</a>
```

Images & media: quality and meaning

Key HTML elements

- <img>: images (always include alt text).
- <figure> + <figcaption>: image with caption.
- <audio> / <video>: built-in media players.
- <source>: provide multiple formats for compatibility.

Example

```
<figure>
  <img src="campus.jpg" alt="Computer science building at sunset" />
  <figcaption> Our department building (2026).</figcaption>
</figure>
```

Best practices

- Use correct formats: JPEG for photos, PNG for transparency, SVG for icons/logos.
- Optimize file size (large images slow pages).
- Responsive images: srcset/sizes for different screens.
- alt text: describe the meaning, not decoration (use alt="" for purely decorative).

```
<audio controls>
  <source src="mmm.m4a"
        type="audio/ogg">
  <source src="horse.mp3"
type="audio/mpeg">
  Your browser does not support the audio
        element.
</audio>
```


Lists & tables: presenting structured information

Lists (most common)

- <ul> unordered list (bullets).
- <ol> ordered list (1,2,3…).
- <li> list item.
- Use lists for navigation menus, features, steps, requirements.

Quick list example

1. Plan the page
2. Write semantic HTML
3. Style with CSS

```
<ol>
  <li>Plan the page</li>
  <li>Write semantic HTML</li>
  <li>Style with CSS</li>
</ol>
```

Tables (use only for tabular data)

- <table>, <tr>, <th>, <td>.
- Good for data like schedules, marks, price lists.
- Not recommended for page layout (old practice).
- Add <caption> and proper headers for accessibility.

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Quick list example

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  <li>Plan the page</li>
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1. Plan the page
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```
<ul>
  <li>Plan the page</li>
  <li>Write semantic HTML</li>
  <li>Style with CSS</li>
</ul>
```

- Plan the page
- Write semantic HTML
- Style with CSS

- A description list is a list of terms, with a description of each term.
- The <dl> tag defines the description list, the <dt> tag defines the term (name), and the <dd> tag describes each term:

Example

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

Coffee
- black hot drink
Milk
- white cold drink